International Energy Agency

Uganda Grid Transmission Investment Plan

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Introduction

Uganda Grid Transmission Financing – Key Area

GIS Files with MCA prioritisation linked to high value areas to be integrated into UETCL website

Historical Funding Analysis amount accessed, areas for potential growth, and comparator countries

Policy Recommendations on how to scale up deployment of the grid

Investment Plan advice on approaches to grid investment, including international case studies

Timeline

Oct 2024
Scoping Mission
Uganda & Kenya to confirm work with gov and meet stakeholders

Feb 2025
Kick off analysis
Start GIS analysis, data gathering and modelling

July 2025
Initial Findings
Present draft findings to Government and stakeholders

Sept 2025
Workshop
Present preliminary results to Government and stakeholders.

Dec 2025
Launch Report
Launch final report

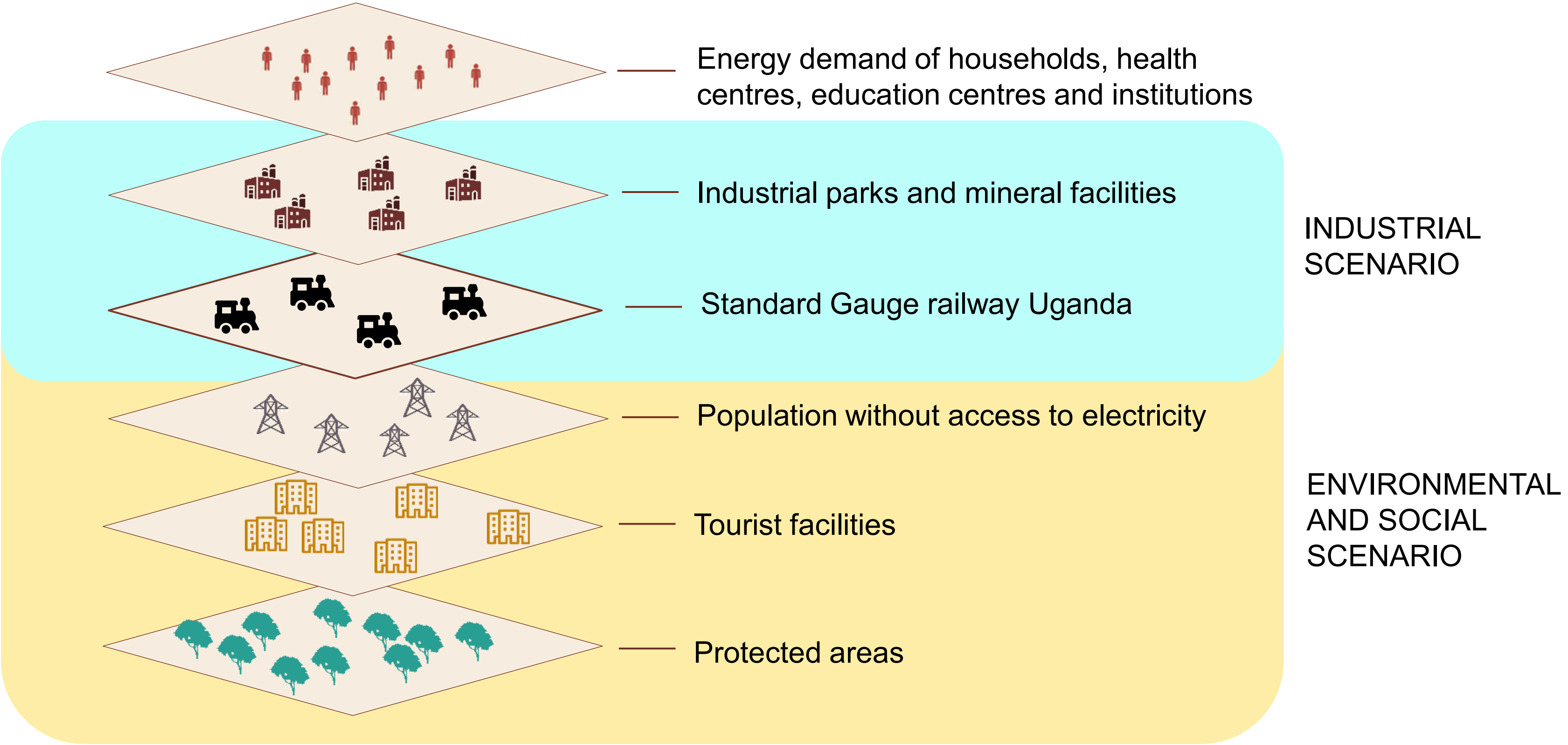
Previous work and references

- Uganda energy policy review
- Uganda energy transition plan
- Uganda electricity access scale up project
- Grid Development Plan 2023-2040
- Fourth National Development Plan (NDPIV) 2025/26 - 2029/30
- Integrated Energy Resource Master Plan (IERMP): Demand forecast report
- Guidelines for developing Uganda's industrial parks and freezone
- Energy Sector GIS Working Group (GIS input files)

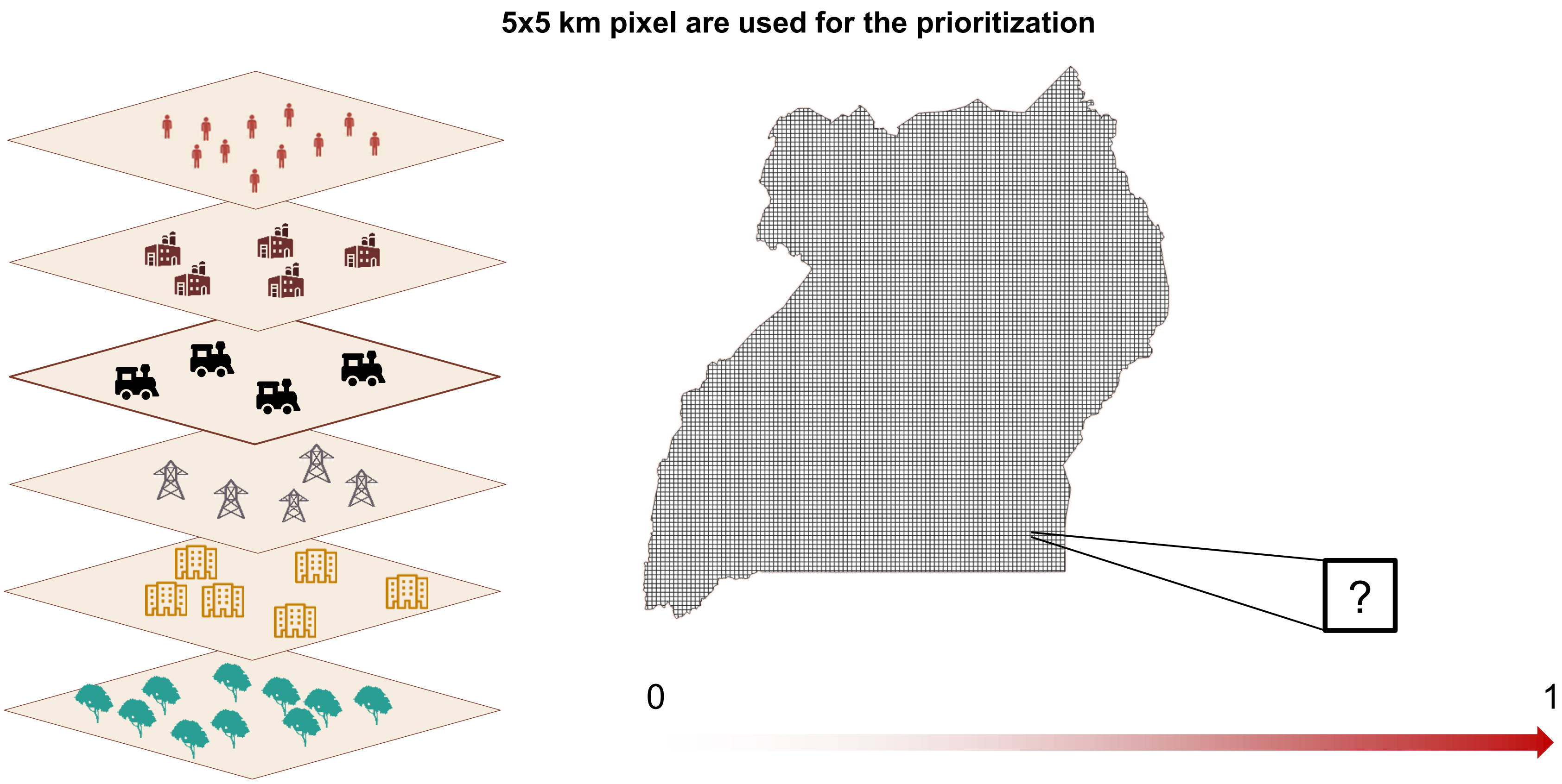
Multi criteria analysis

Different layers are considered, and scenarios are modelled

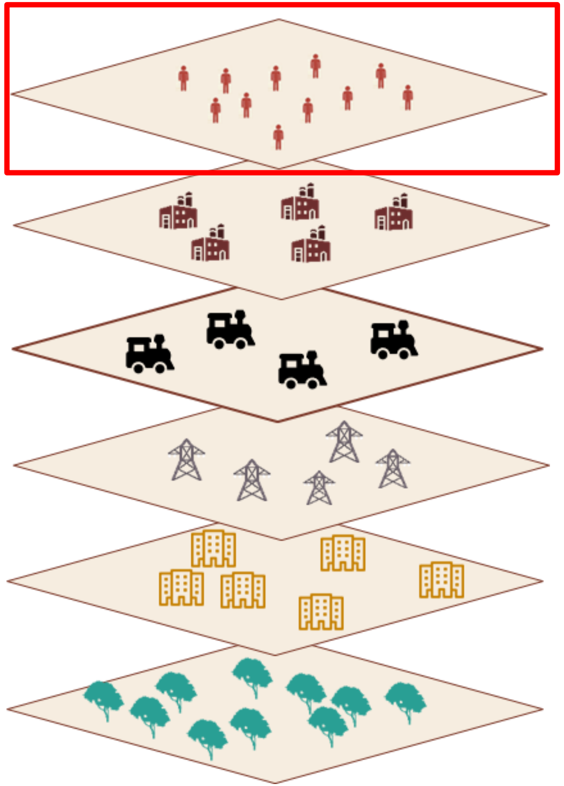
Modelling approach for electrification prioritisation



Different layers are considered, and scenarios are modelled



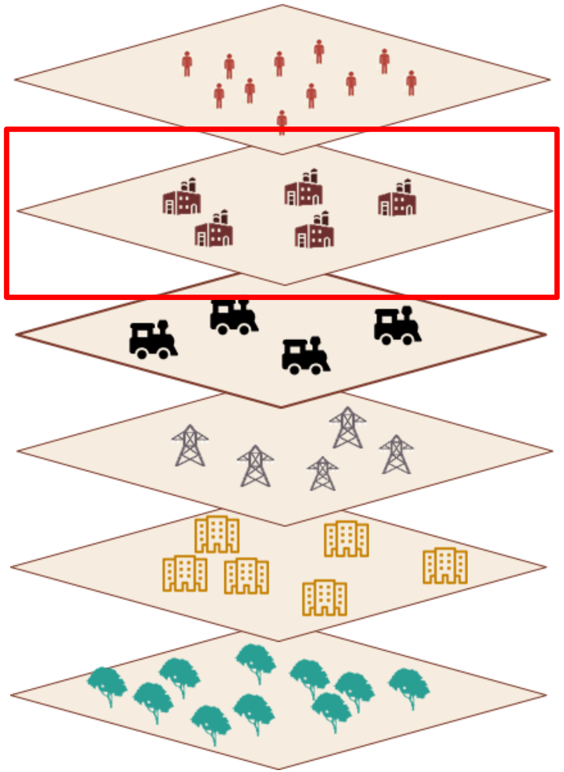
… based on the ATMS scenario for economic development



Energy demand of households, health centres, education centres and institutions

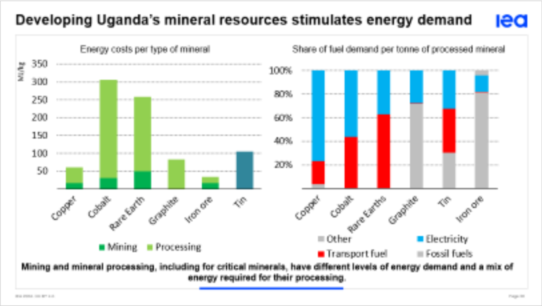
- around 3kWh/day per household (coherent with national objectives)

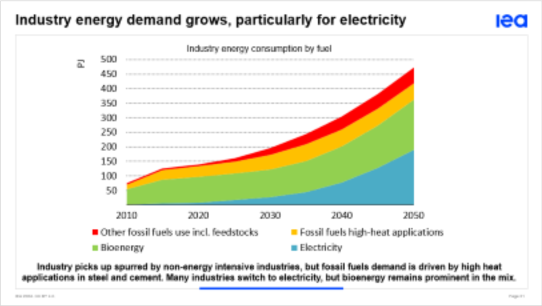
… mineral facilities and industries as basis of economic development



Energy demand of households, health centres, education centres and institutions

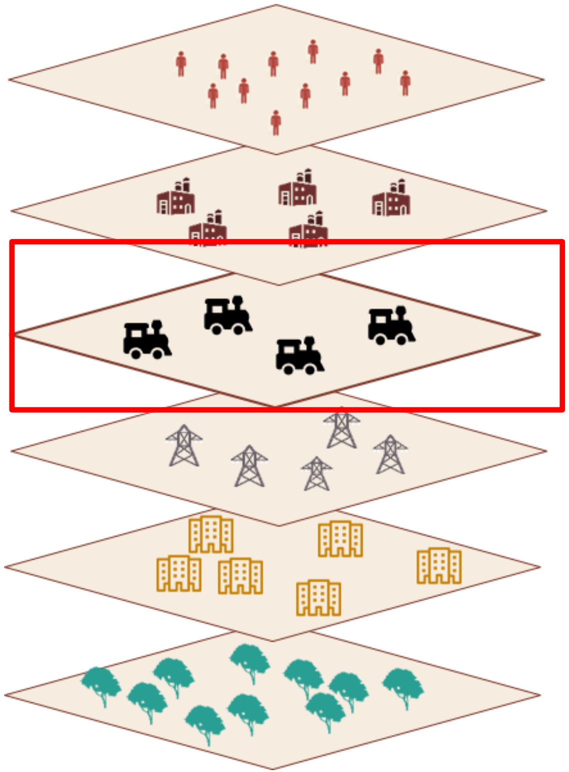
Mineral facilities and industrial parks





What should we prioritise ?

… that required strong and spread connections

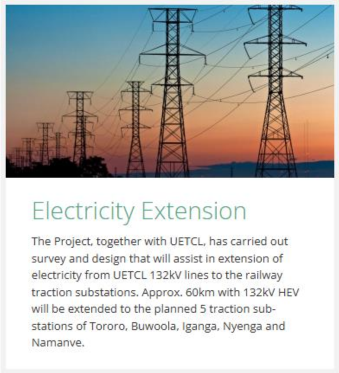


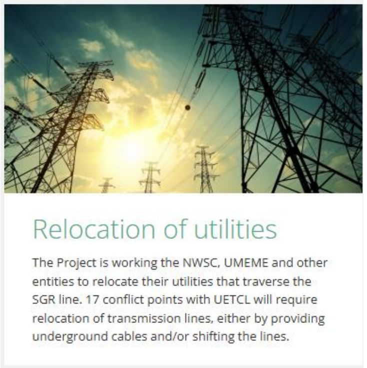
Energy demand of households, health centres, education centres and institutions

- around 3kWh/day per capita (coherent with national objectives)

Mineral facilities and industrial parks

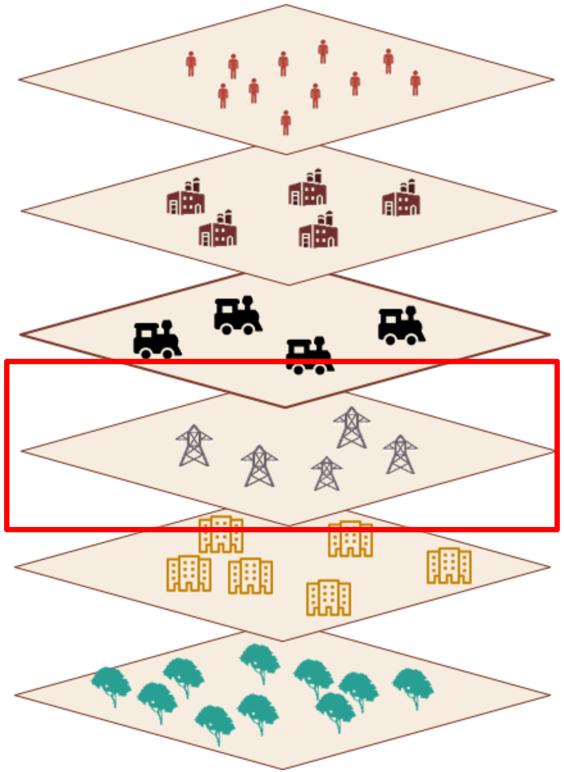
Standard Gauge railway Uganda





Is there anything more we should include in the analysis ?

However, 60% of population does not have access to electricity



Energy demand of households, health centres, education centres and institutions

- around 3kWh/day per capita (coherent with national objectives)

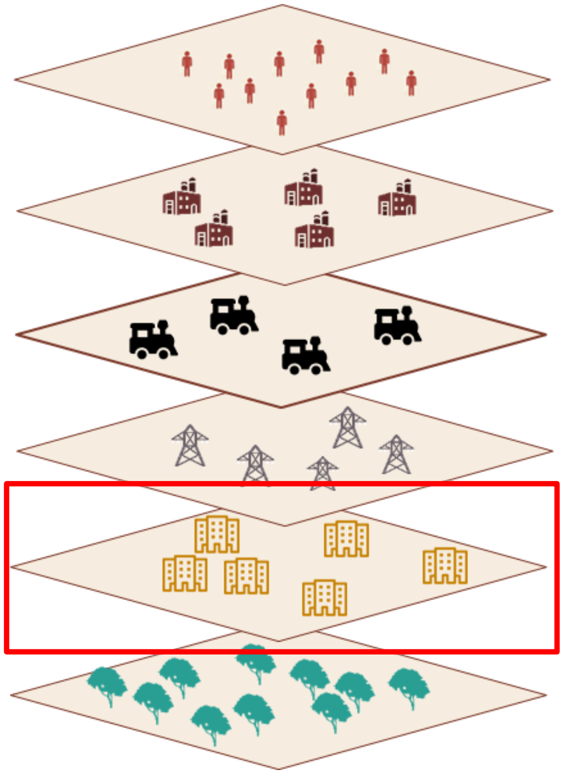
Mineral facilities and industrial parks

Standard Gauge railway Uganda

Population without electricity

- Non quantitative indicator to be prioritised in the social scenario

Tourism is also considered a pillar for development



Energy demand of households, health centres, education centres and institutions

Mineral facilities and industrial parks

Standard Gauge railway Uganda

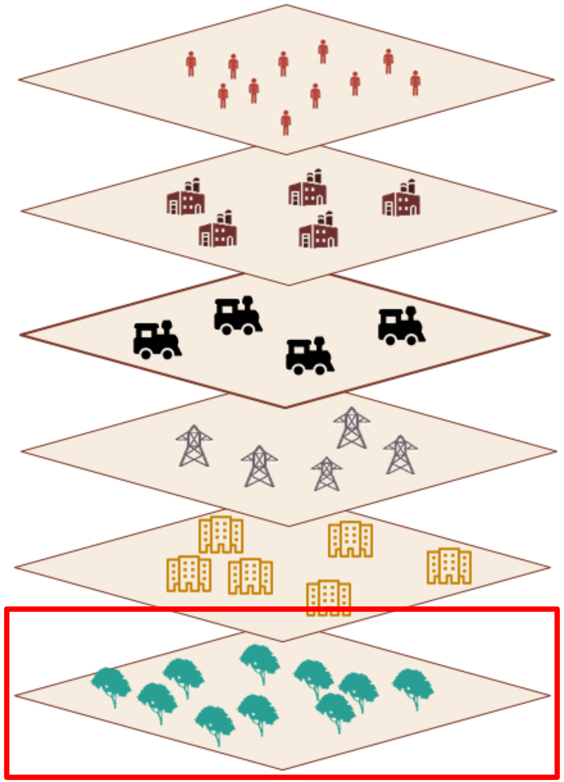
Population without electricity

Touristic hubs and centres

- ONGOING

Is there an investment plan for the tourism sector ?

Uganda is home to national parks and wildlife reserves



Energy demand of households, health centres, education centres and institutions

Mineral facilities and industrial parks

Standard Gauge railway Uganda

Population without electricity

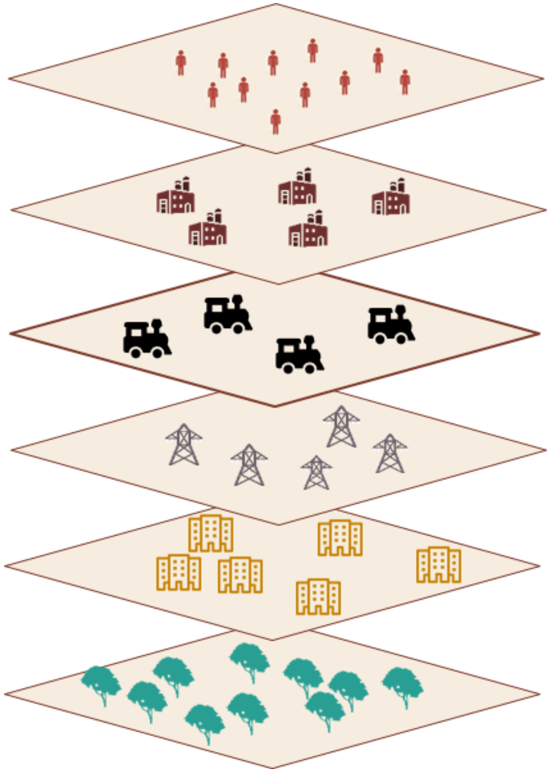
Touristic hubs and centres

- ONGOING

Protected and natural areas

- Non-quantitative indicator to be included

Finally, the multicriteria analysis is based on scores



Scenario

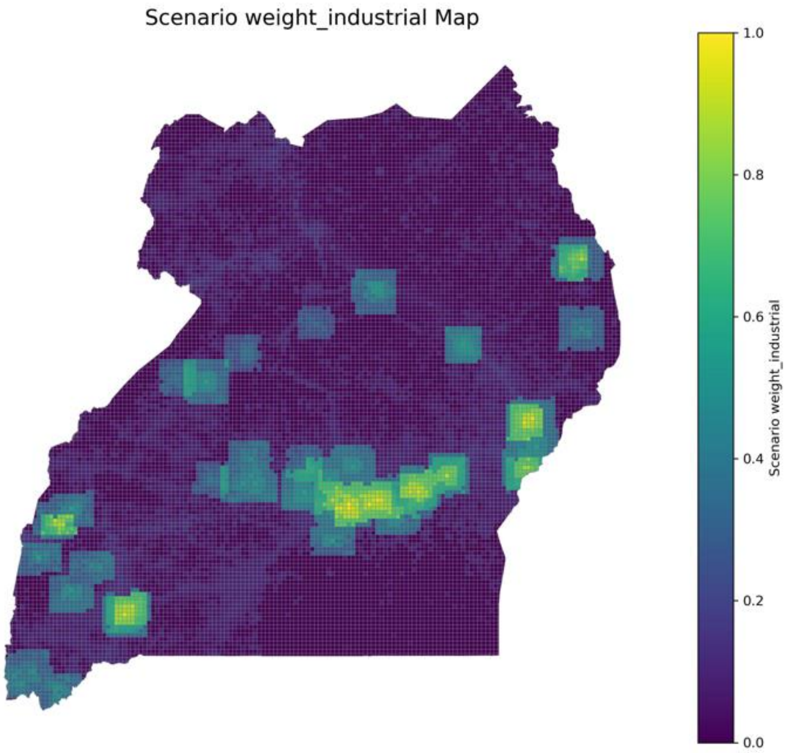
- Right now, just one scenario as example
- All the score above are multiplied by a weight and summed u

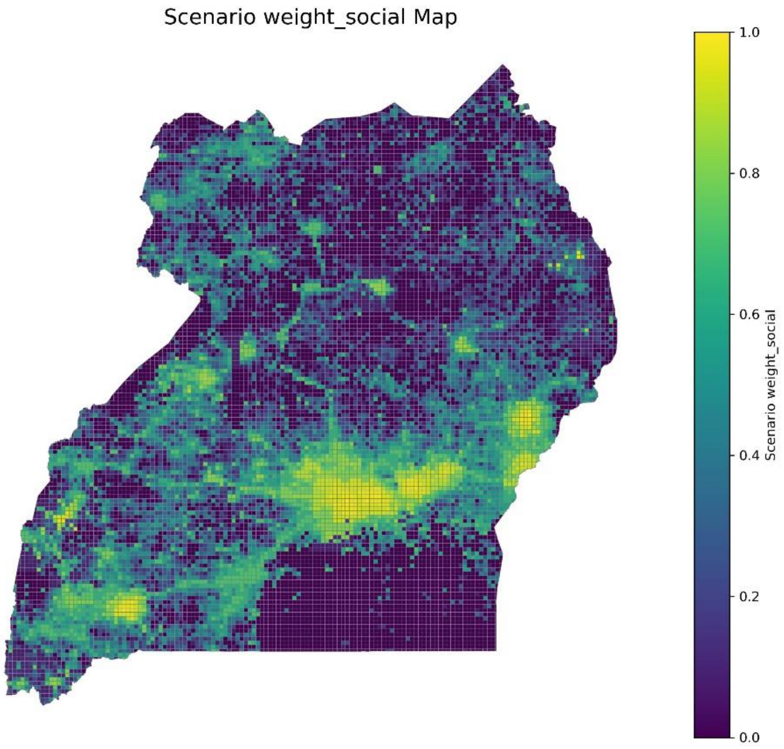
Final_score = sum(Score_i * weight_i)

Industrial and social scenarios ? Others ?

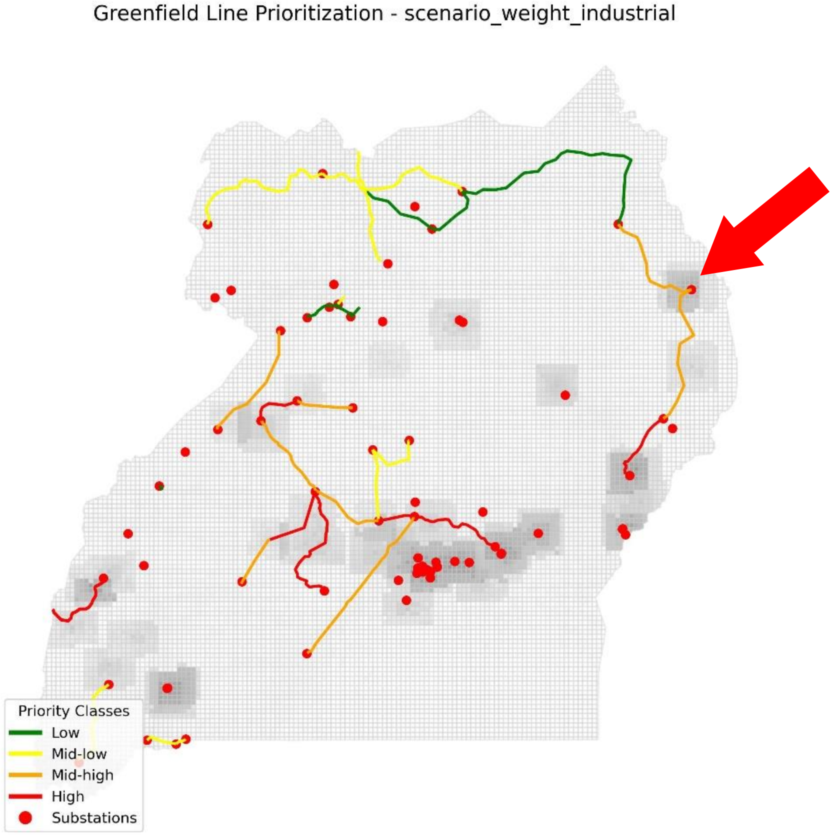
Transmission lines prioritisation

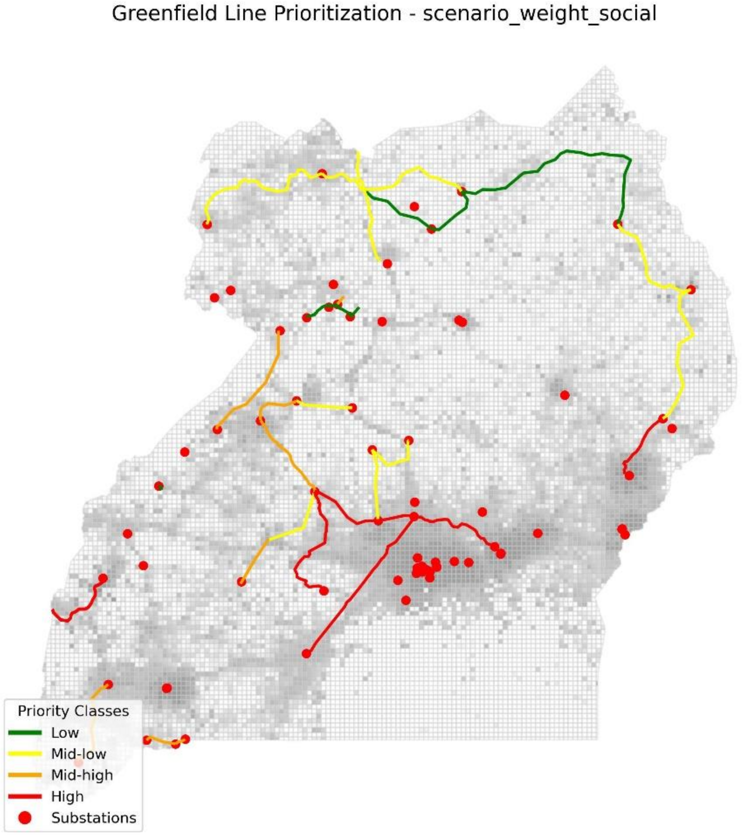
Uganda Transmission prioritisation



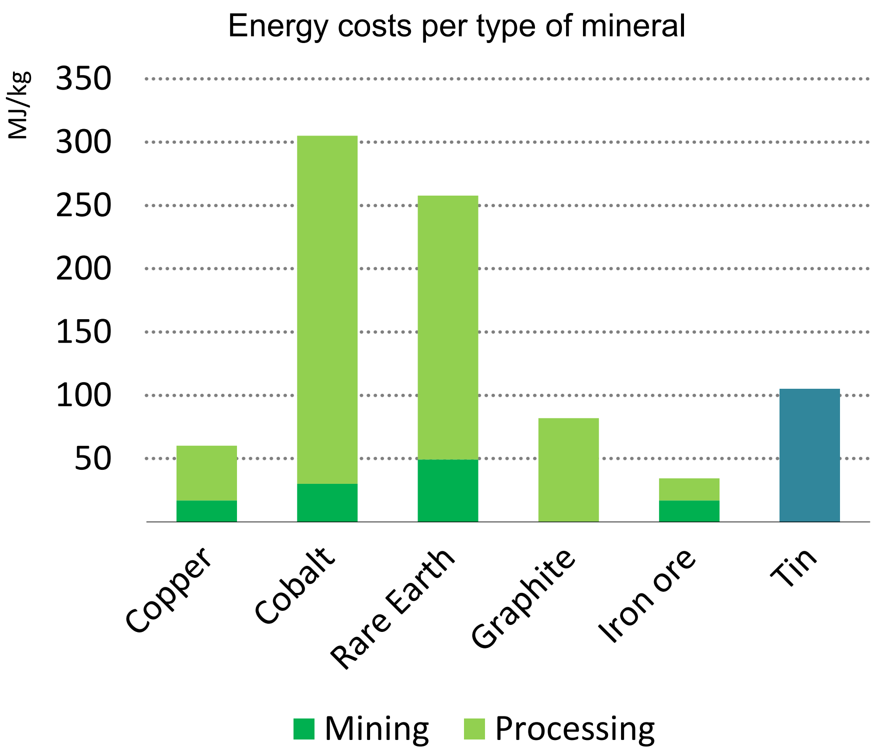


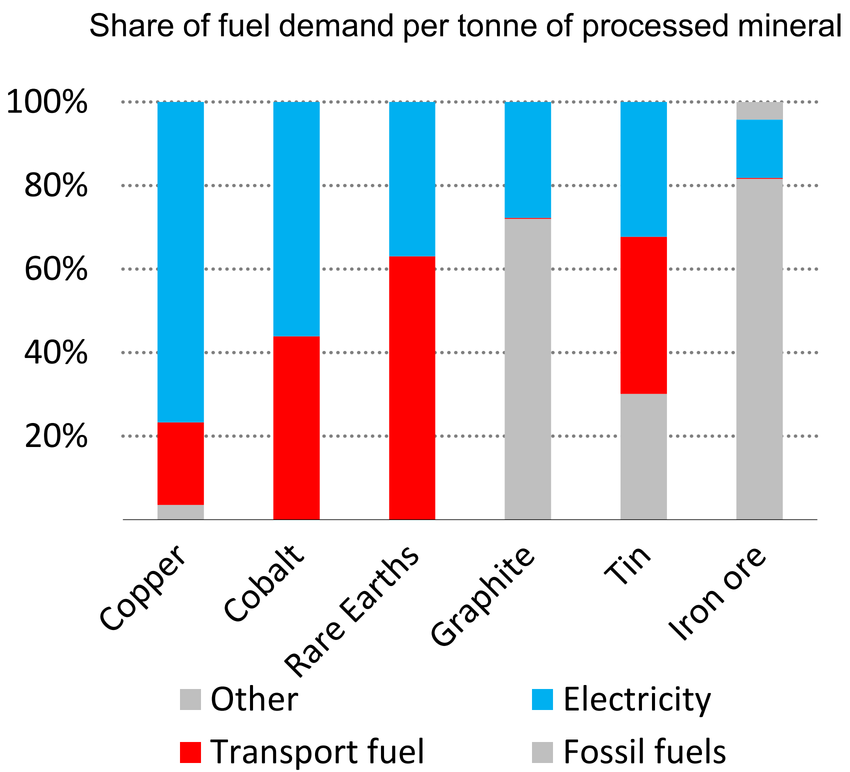
Uganda Transmission prioritisation





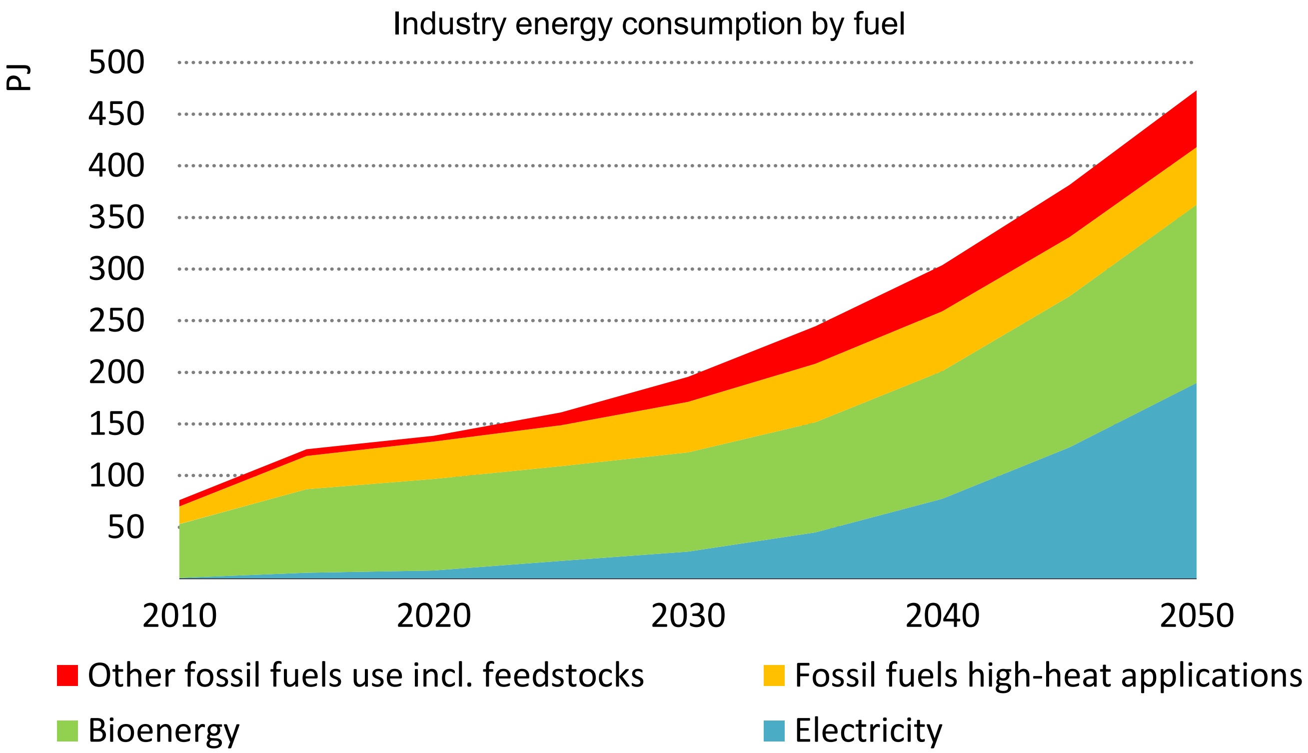
Developing Uganda’s mineral resources stimulates energy demand





Mining and mineral processing, including for critical minerals, have different levels of energy demand and a mix of energy required for their processing.

Industry energy demand grows, particularly for electricity



Industry picks up spurred by non-energy intensive industries, but fossil fuels demand is driven by high heat applications in steel and cement. Many industries switch to electricity, but bioenergy remains prominent in the mix.